

How Internet of Things can influence the sustainability performance of logistics industries – A Chinese case study

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ABSTRACT

The introduction of new information technologies has started reshaping global industrial sectors and supply chains. Due to the introduction of the Internet of Things (IoT) real-time management strategies have been adopted by the global logistics industry, turning the branch into an intelligent service supplier. This paper assesses the influence of IoT on the Chinese logistics sector and related environmental performance between 2011 and 2018. This paper establishes an evaluation framework for the logistics performance under social, economic and environmental dimensions by using time entropy weighting. Using a grey correlation approach, we identify a strong positive correlation between the logistics performance and the IoT market scale. We further find a significant and increasing correlation trend for an expansion of the IoT market and the reduction of carbon and PM 2.5 intensity. The environmental regulation though positively correlated with logistics sustainability, shows less potential to directly improve economic and social performance. It also indirectly promotes sustainability performance of the logistics industry through support for technological innovation. High investment in IoT is estimated to limit the potential of small and medium-sized enterprises to increase their labor compensation and expand the scale of employment. Finally, we project China's IoT market developments for 2021–2025 using a grey forecasting model considering the influence of investment confidence and COVID-19. The results indicate that China's share of the global IoT market will likely rise from 18% to 30% by 2025.

1. Introduction

Since the beginning of the industrial revolution, a number of key innovations such as the steam engine, electrification and the internal combustion engine changed the way we produce and deliver goods (Edquist and Magnus, 2006). Such innovations enabled a higher level of flexibility, increased the overall productivity (Smil, 2000; Schurr, 1984) and resulted in a reorganization of the industrial production structure (Devine 1983). Due to the availability of mechanical power, we nowadays see a highly complex global structure of supply chains (Baldwin, 2011; Koopman et al., 2014; Coe et al., 2008). Ideal production facilities and connections are determined by a myriad of factors. New approaches such as just-in time production, lean logistics, and vendor managed inventory have become the norm (Coe et al., 2008). They contribute to an efficient, relatively low-cost sustainable supply chain network, which at the same time has vulnerabilities since redundancy and buffers in

inventories usually are low (Acemoglu et al., 2012), as recently seen in the Covid crisis (Gultekin et al., 2022).

Recently, advanced information and communication technology (ICT), such as the IoT has gained market shares under the development of Radio Frequency Identification (RFID), Manufacturing Resources Planning (MRP) and Enterprise Resource Planning (ERP). It will likely trigger another refinement of how things are produced and how production capacities will be distributed across the world (Sarkar et al., 2018). The significant increase in the number of connections between machines, the forecasted market expansion, higher data availability and new optimization options for production and resource allocation indicate that an early impact of the IoT can be observed in the manufacturing and logistics industries (Manyika et al., 2015).

Scholars highlighted that IoT will trigger unprecedented environmental and social developments. Ali and Shahram (2020) assessed a potential carbon emission performance improvement of IoT through

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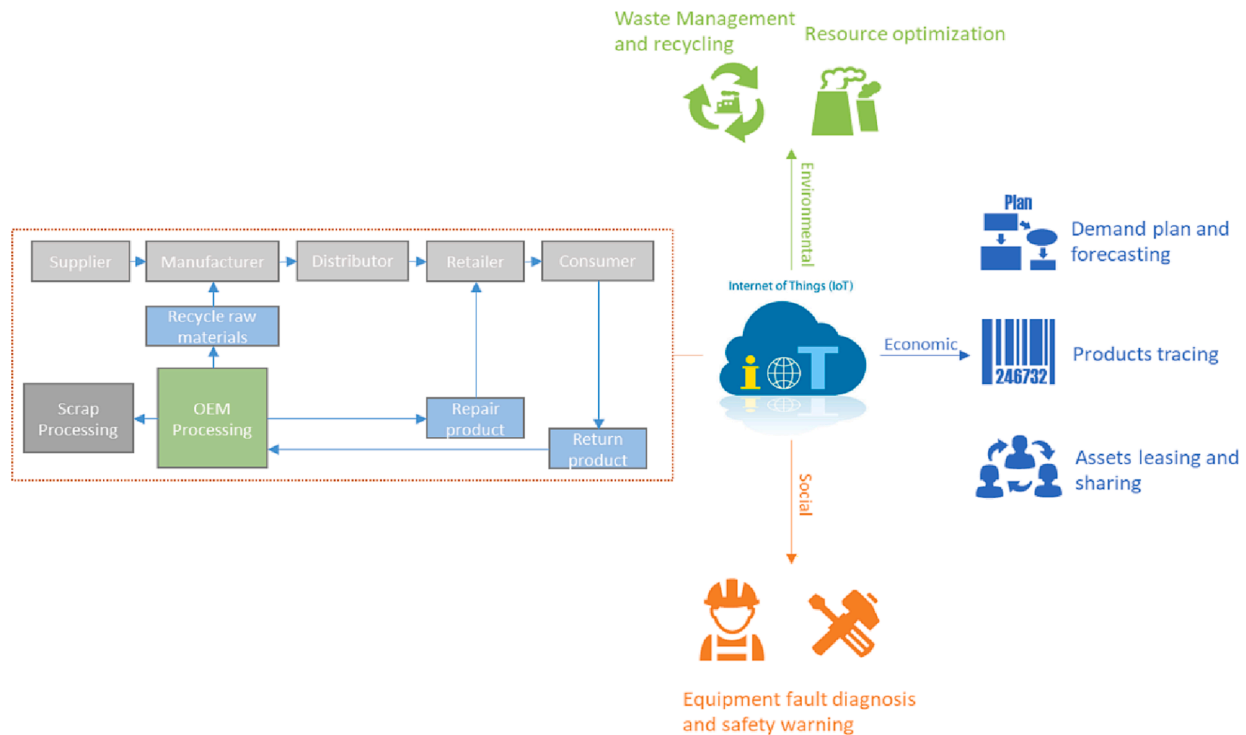


Fig. 1. IoT enabled green logistics under TBL dimensions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

hardware, software, communication and architecture in smart home and industrial control processes. Pan et al. (2020) showed that the implementation of smart logistics could curb carbon emissions and might help to achieve a sustainable society.

As the largest emitter of greenhouse gases, China listed the IoT as one of the seven strategic emerging industries¹ (MIIT, 2012). IoT connections have been growing fast in China (GSMA, 2020). The number of IoT vendors from China has increased from 16 % in 2019 to 22 % in 2021 (Wegner, 2021). Similar dynamics also occurred in the rapidly growing Chinese logistics industry (GOSC (General Office of the State Council, PRC). (2014)),² where the bike sharing 4.0 system supported by the IoT contributed to reducing carbon emission, as it conveniently solves the problem of the “last mile” in logistics³ (Chen et al., 2020). However, Alcayaga et al. (2019) mentioned that if IoT based smart systems could trigger a comprehensive sustainability performance in logistic industry, is still largely unknown.

Given these dynamical changes this paper addresses the following research questions with a focus on the development of China’s logistics industry and its performance under consideration of environmental, economic and social aspects over the period 2011–2018. The analysis is based on an innovative grey correlation analysis:

- i) How can the sustainability performance of the logistics sector be measured?
- ii) How has IoT been correlated with the sustainability performance of China’s logistics sector?
- iii) What is the market prospect of IoT considering the investment confidence uncertainties and what does it imply for future sustainability performance?

The contribution of this paper are as follows: First, this study designs an evaluation framework to identify the role of IoT and other factors in improving the Triple Bottom Line (TBL) performance of the Chinese logistics sector. Second, we develop an improved entropy weight approach that takes time factor into account to calculate weighted TBL performance. Third, we apply a grey relational model to analyze the correlation effect between IoT development and the weighted TBL dimensions of the logistics industry. We assess further the role of environmental regulation as a contributor to the TBL performance. Finally, we use the grey forecasting method to estimate the China’s IoT market prospects in logistics under different logistics prosperity index (LPI) scenarios.

The structure of this article is as follows. Section 2 introduces the analytical framework. Section 3 indicates the mathematical models. Section 4 presents and discusses the results. Finally, section 5 reflects on the key findings, lists the limitations and proposes the future steps.

2. Analysis – Sustainability performance evaluation framework

2.1. Strategies for improving sustainability performance of logistic industry with IoT

The General Office of the State Council of the PRC stated that the logistics industry needs to be high value-added and low carbon emissions to realize until now untapped sustainability performance potentials (GOSC (General Office of the State Council, PRC). (2014)). This sustainable logistics vision also considers circular economy (CE) practices and applies information technology including IoT into it to

¹ Seven strategic emerging industries contain energy saving and environmental protection industry, emerging information industry (mainly include IoT and next generation network), biological industry, new energy industry, new energy vehicles industry, high-end equipment manufacturing and new materials industries.

² As of 2019, the added value of China’s logistics industry accounted for 7.15% of GDP, with an average annual value-added growth rate of 8%, and the proportion of logistics costs has dropped from 18% to 14.2%.

³ Bike sharing 4.0 system demands no fixed station and all bikes are connected to the Internet, it can be used through scanning the QR code with a smartphone of its body.

minimize unnecessary waste and achieve traceability through whole supply chains (Sasikumar and Kannan, 2008a; Sasikumar and Kannan, 2008b; GOSC (General Office of the State Council, PRC). (2014); Husain and Malik, 2020), as visualized in Fig. 1.

It depicts IoT's role into environmental, economic and social dimensions following the concept of the TBL. For instance, there is indication that IoT can help reducing 20–30 % energy use and CO₂ emission in industrial value chains via monitoring, tracking, optimization and design evolution under TBL dimensions (Ding, 2022).

The IoT can increase the profitability through automated/semi-automated operations, preventive maintenance of monitoring machines and robots, and continuous communication (Roy and Roy, 2019). For logistics sector, it is used to obtain real-time information of the position of vehicles and goods by the logistics enterprises. Then, the acquired logistics information is transmitted back and processed to help decision making. The real-time information sensing and processing system uploads the real-time task information of vehicles to enterprises information integrated system⁴ (EIS). With the help of Geographic information system (GIS) current road condition is monitored which supports path optimization algorithms to provide and visualize the optimal transport routing (Liu et al., 2018). The adoption of novel Internet-based transactions can exploit information for a faster and more sustainable collection of waste from whole supply chain, leading to lower environmental costs associated with the recovery process (Al-Masri et al., 2018; Garrido-Hidalgo et al., 2020). These features enable the IoT to improve profitability, reducing unnecessary emissions.

IoT can also contribute to a safer environment for workers and make value chains and business models more transparent for business partners (Roy and Roy, 2019). This enhanced transparency encourages ethical behavior among the business stakeholders and collaborators, thus increasing the reputation of industries and strengthening cooperation in the logistics industry (Tan et al., 2020).

However, Esmailian et al. (2020) believed that a new evaluation framework at the industrial level should be established to assess the sustainability consequences of the IoT in green logistics. Therefore, we first develop an indicator framework to assess the logistic industry's sustainability performance under IoT interventions.

2.2. Indicators in the evaluation framework

Green logistics depicts the management of material, information and capital flows as well as cooperation among companies along the supply chain while taking goals from TBL dimensions into account (Seuring and Müller, 2008; Carter and Rogers, 2008). Dubey et al. (2015) suggested supplier relationship management (SRM), reduction in pollution and CO₂ emission (RE), increase in full-time equivalent employees and compensation (FEEC) and in value added (VA), green technology adoption (GTA), total quality management (TQM) as main enablers of green logistics practices.

2.3. Total quality management (TQM)

TQM is a widely used tool for quality and operating cost control. Jabbour et al. (2014) viewed quality management as a sign of maturity in economic management, therefore it likely influences the green logistics performance. Brkić et al. (2016) further concluded that those firms implementing TQM have achieved superior total factor productivity (TFP). As we are interested in three different underlying dimensions, we split the TFP in Labor productivity (LP, value added per persons) measured as VA generated per year per full time equivalent persons employed, and Capital productivity (CP, value added per unit of

Table 1

Evaluation of Logistic Industry's Sustainability Performance.

Topic Layer	Explanation Layer	Indicator Layer
Economic	VA TQM	Value-added (VA) Labor productivity (LP) Capital productivity (CP)
Environmental	GTA RE	Contribution of green-tech patents (CGP) Carbon intensity (CI)PM 2.5 intensity (PI)
Social	FEEC SRM	Employment population (EP)Labor compensation (LC) Total investment (TI)

logistics expenses capital) measured as VA generated per year divided by the annual expenses (Including transportation, storage, and management expenses) in logistics. LP and CP explain labor and capital inputs of TFP respectively (Zimon, 2016; Putri et al., 2017).

2.4. Supplier relationship management (SRM)

Vachon and Klassen (2006) illustrated the value of collaborating with suppliers for green logistics. In particular they argued that inter-organizational collaboration particularly with suppliers, contributes to sustainability performance. Wagner and Bode (2014) argued that suppliers that make relationship-specific investments are more likely to sustain or even increase the business activities with the corresponding customers. A supplier seeks to earn the returns on its investment and is therefore interested in sustaining a long-term relationship with the corresponding buying firm. This will result in stronger and closer integration of the two firms and reduced operating and purchasing costs from both sides. Besides, these investments help to improve product quality and reduce environmental costs in production (Kang et al., 2009). Due to the difficulty of obtaining specific data on relationship-specific investments in the logistics industry. Scholars showed that fixed assets investment (FAI) and foreign direct investment (FDI) in logistics can be used to approximate the SRM effect and green transformation (Mao et al., 2019). They promote technology diffusion by imitation learning, human and capital flow to enhance supplier relationship and achieve cleaner production.

2.5. Green technology adoption (GTA)

The environmental progress of the logistics industry is reflected in the contribution of green-tech patents (Kang et al., 2021). Sikdar and Howell (1998) presented the importance of green technology to reduce the negative impacts of industrial activities on the planet through a review of cleaner unit process development. Most green-tech firms in logistics in China are innovative small and medium-sized enterprises (SMEs). Therefore, it is difficult to identify their research and development (R&D) expenses for specific green technologies. Nevertheless, the number of green-tech patents can be used as a proxy to measure the GTA (Marra et al., 2017).

2.6. Reduction in pollution and CO₂ emission (RE)

Part of the sustainability performance of the logistics industry is reflected in reductions of environmental burdens, such as carbon emissions and fine particles emissions. These two dimensions are core considerations of the 11th –13th Five Year Plans in China (Price et al., 2011; Hu, 2016). Carbon emission reduction is also regarded as one of the key enablers of greener logistical practices (Dubey et al., 2015) and shows a correlation with energy conservation and emission reduction of

⁴ Systems integration refers to the process of linking together all the different computer systems and software applications so that they can function as a complete system.

Table 2

Time entropy weight from 2011 to 2018.

Time	2011	2012	2013	2014	2015	2016	2017	2018
Weight (γ_t)	0.0228	0.0335	0.0491	0.0721	0.1058	0.1552	0.2276	0.3339

logistics industry (Quan et al., 2020). PM 2.5 is a key air pollutant that poses great risk to health. The logistics sector is a major contributor of PM 2.5 emissions (Zhang et al., 2021).

2.7. Increase in Value-added (VA)

The green logistics practices and an increased focus on the circular R's (i.e., reduction, reuse, repair, recycling, remanufacture and redesign) can help organizations to improve profitability while at the same time increase market shares through enhanced customer satisfaction and improved sustainability performance (Dubey et al., 2015). Value-added reflects the value generated by producing goods and services, and is measured as the value of output minus the value of intermediate commodity inputs. It also represents the income available in the production process and can directly reflect the economic performance of the logistics industry (Gabrielli et al., 2020).

2.8. Increase in full-time equivalent employees and compensation (FEEC)

Ahi and Searcy (2013) reviewed the definitions for green logistics and sustainable supply chain management. They find indication that green logistic performance is directly related to firms' aims of achieving a greater market size. As firms have increasingly embraced better performance on TBL dimensions, they have shifted their focus toward creating value for all stakeholders impacted by business decisions, including customers, employees, and community members (Pfajfar et al., 2022). To achieve this goal, while having higher added value, enterprises also need to create more jobs opportunities and provide higher salaries for employees (Mariappanadar, 2003). Insufficient investment in employees will cause a negative impact on TBL, especially in social dimension (Alvaro and Ramon, 2020).

Therefore, a comprehensive evaluation index which contains 9 quantitative indicators was discerned in Table 1 explaining these main enablers of green logistics practices.

The evaluation framework also describes the three TBL dimensions. As for the economic dimension, LP is an indicator that is closely linked to economic growth, competitiveness, and living standards within an economy. CP is used to measure the efficiency of logistics expenses which contains shipment, storage and administrative expenses (LSY, 2019). As shown in formula (1), where AV_L^t represents added value of logistics industry at time t , N_L^t represents number of full-time equivalent labors in logistics sector, EXP_L^t represents total capital expenses in logistics sector, LP_t represents labor productivity at time t , CP_t represents capital productivity at time t .

$$LP_t = \frac{AV_L^t}{N_L^t} \times 100\% \quad (1)$$

$$CP_t = \frac{AV_L^t}{EXP_L^t} \times 100\%$$

Statistically, we used the ratio of the patents on green technologies OP_g of logistics to the total output patents OP_t to present the GTA effect, as shown in formula (2). The greater the ratio, the greater the contribution of green-tech patents. This indicator positively describes the contribution of green technology into industrial activities, and present the advancement of logistics in sustainability performance to some extent.

$$CGP_g = \frac{OP_g}{OP_t} \times 100\% \quad (2)$$

The total emissions of CO₂ and PM 2.5 are represented by EM_c and EM_{pm} , respectively. CI represents carbon emission per unit of added value. PI represents PM 2.5 emission per unit of added value. Both CI and PI are shown in formula (3)

$$CI = \frac{EM_c}{AV_L} \quad (3)$$

$$PI = \frac{EM_{pm}}{AV_L}$$

The last component of the TBL highlights the societal impact of industries, or their commitment to people. EP (the number of full-time equivalent employees in logistics sector) and LC (the total compensation of full-time equivalent employees in logistics sector) reflect these values from social aspect (Li and Zou, 2018). In addition, domestic and international investment help to accelerate the rapid expansion of China's domestic demand. More than half of the world's new manufacturing and logistics jobs were created in China since the global financial crisis (Abdelaaziz and Uri, 2019). It also helps to form corporate social responsibility with other organizations that share a common purpose-driven goal (Pfajfar et al., 2022). These two values are summed up to describe total investment (TI).

The indicator data related to national logistics used in this manuscript come from the 2012–2019 China Statistical Yearbook (CSY, 2012–2019) and 2012–2019 China Logistics Yearbook (CLY, 2012–2019). The CO₂ and PM 2.5 emission datasets of China's logistics industry come from IEA and MEIC database, respectively. (IEA (International Energy Agency) (2020); Li et al., 2017; Zheng et al., 2018). Patents on environment technologies datasets are provided by the OECD (OECD, 2021). We select information transmission, software and information technology service industries in the statistical yearbook as statistical indicators of high-tech industry in logistics. Information on the market size of the IoT in China is taken from "2018–2019 The Annual IoT Industry Development Report of China" (CEIS, 2019). Due to the lack of specific data on the proportion of the Chinese logistics IoT market, we approximate China's logistics industry share through using the global proportion of the IoT market size in the logistics industry (Wegner, 2021). See Appendix B.

3. Mathematical models

The mathematical models are connected through multiple steps. First, based on the evaluation framework in section 2, a time entropy weight approach is used to weight each factor (Qian et al. 2016; Geng et al. 2019). After a standardized processing, the measurement results

Table 3

Indicators weights for logistics industry evaluation index.

Topic Layer	Indicator Layer	Weight (%)
Economic	VA	11.21
	CP	11.27
	LP	11.00
Environmental	CI	12.84
	PI	9.57
	CGP	12.11
Social	EP	10.70
	LC	10.67
	TI	10.63

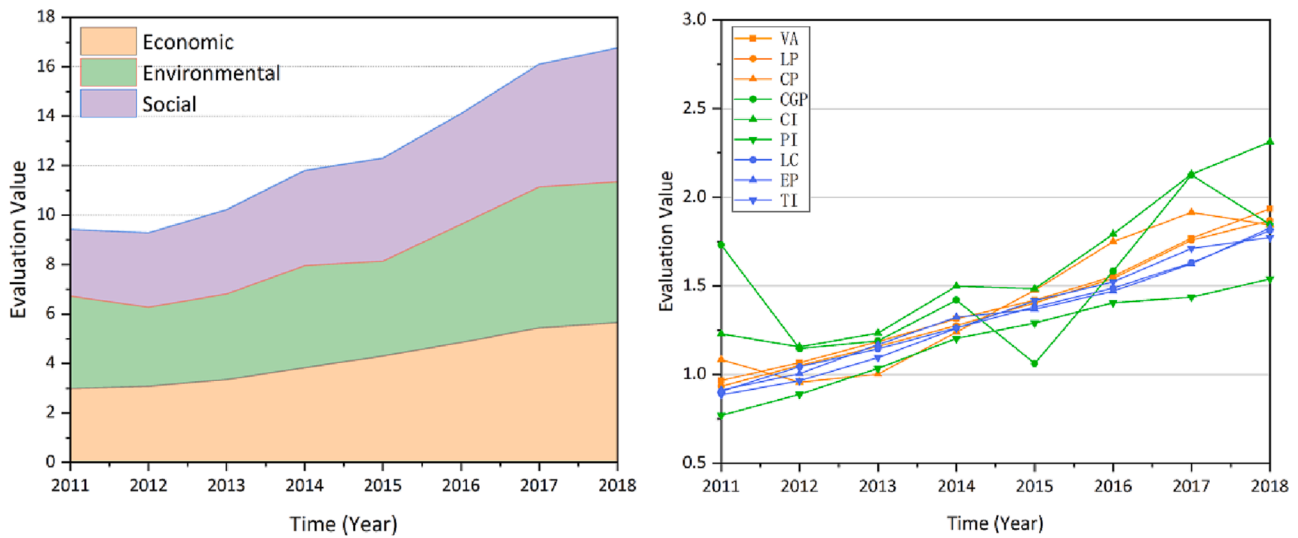


Fig. 2. Sustainability performance of the logistics industry evaluation value between 2011 and 2018.

for logistics industry sustainability performance in each year are calculated. Subsequently, considering the potential impact of environmental policies on the logistics industry sustainability performance, the grey relational degree model (Wei and Xie, 2019) is applied to analyze the relation degree between the IoT market, environmental regulation and sustainability performance of logistics industry, respectively. Finally, the exogenous variable expanded grey forecasting method (Liu et al. 2019) is used to forecast the development of IoT market.

3.1. Method to calculate entropy weight of each factor

In information theory, entropy is a measure of uncertainty. The greater the amount of information, the higher the certainty and smaller the entropy value. Therefore, the degree of dispersion of a certain indicator can be judged by calculating the entropy value. The greater the degree of dispersion of the indicator, the higher its influence on the comprehensive evaluation. As Weng et al. (2020) indicated, the approach described the uncertainty, and it provides the guides for the research on the nonlinear theory.

The Entropy Weight Coefficient Method (EWCM) is one of the methods which applies entropy theory. Wan (2009) made a comparison between this method and the traditional factor analysis method, which reflect the same evaluation result in the research of a city's comprehensive competitiveness. Gao et al. (2017) synthesized and compared diverse subjective (Fuzzy AHP) and objective (Entropy and CRITIC) weighting methods to evaluate the four candidate options of nuclear fuel cycles in China based on environmental, economic, and social considerations. It showed that the integrated and the objective weighting method have similar weight values for each indicator. While the subjective weighting method differs significantly due to conflicting criteria among the competing decision makers at different levels of expertise and experience. EWCM overcomes the subjectivity of the evaluation and the problems of the internal factors that affect the sustainable development system compared with other evaluation methods, and the integrity of evaluation results was also taken into consideration (Wang et al., 2019). Therefore, this approach has been applied in multiple measurement systems including urban carrying capacity (Weng et al., 2020), environmental conflict analysis, and the sustainable development capability (Li et al., 2019).

However, traditional EWCM ignores the importance of time weight (Guo et al., 2007), which reflects the importance of data at different time points. According to the information priority principle of "emphasis on new information rather than old information" in grey system theory (Deng, 1984), a grey entropy weight method that combines the idea of

grey theory and entropy theory was established to determine the weights at different time points (Guo et al., 2007), as shown in formulas (4) to (7).

$$L = - \sum_{t=1}^p \gamma_t \ln \gamma_t \quad (4)$$

γ_t with entries $(\gamma_1, \gamma_2, \dots, \gamma_p)^T$ represents the time weight vector at different times t , p represents number of years, and L represents the entropy of the time weight vector.

$$\theta = \sum_{t=1}^p \frac{p-t}{p-1} * \gamma_t \quad (5)$$

θ in formula (5) represents the time degree. It reflects the average degree of information contained in the weights in the time sample, and is expressed as a judgment of the importance of each time point in the sequence. In case of $\theta = 1$, $\gamma_t = (1, 0, \dots, 0)^T$ the evaluator attaches the greatest importance to the starting point information. On the opposite $\theta = 0$, $\gamma_t = (0, 0, \dots, 0)^T$ refers to the situation where the evaluator attaches greatest importance to the information of the newest time point. For $\theta = 0.5$, $\gamma_t = (\frac{1}{p}, \frac{1}{p}, \dots, \frac{1}{p})^T$, each point in time t is given an equal importance. Based on the information priority principle of "valuing new information and weakening old information", the value of θ is specified to 0.25 in this paper.

According to formula (5), the solution of γ_t is determined based on θ . The following non-linear programming is constructed with the maximum entropy value as the objective function to solve the time weight of each time point t , which is shown in formula (6):

$$\text{Max} \left(L = - \sum_{t=1}^p \gamma_t \ln \gamma_t \right) \quad (6)$$

The constrains are the following. The sum of all weights is 1, and the weights of adjacent years are always greater than the weights of distant years.

$$\theta = \sum_{t=1}^p \frac{p-t}{p-1} * \gamma_t \quad (7)$$

$$\sum_{t=1}^p \gamma_t = 1$$

$$\gamma_t < \gamma_{t+1}$$

Table 4

Grey correlation degree of the IoT market and three topic layers in 2 periods.

Sequence 1	Sequence 2	Grey correlation degree (2011–2014)	Grey correlation degree (2015–2018)
IoT market	Economic	0.6946*	0.8999**
	Social	0.7666*	0.8488*
	Environment	0.6633*	0.9464**
	Overall	0.6708*	0.9326*

Note: The correlation degree of very high is shown as **, and high is shown as *.

After that, the data need to be standardized as otherwise the measurement units of the collected original would be inconsistent. The standardization process of positive indexes is as follows:

$$x_{ij}^* = \frac{(x_{ij} - \min(x_j)) * \gamma_t}{\max(x_j) - \min(x_j)} \quad (8)$$

Ideally, CI and PI should be as small as possible. Such indicators are called negative indicators and have to be processed and converted to positive values as follows:

$$x_{ij}^* = \frac{(\max(x_j) - x_{ij}) * \gamma_t}{\max(x_j) - \min(x_j)} \quad (9)$$

Where x_{ij} denotes value of the indicator j at time t , x_{ij}^* denotes its normalized value with time weight characteristics, $\max(x_j)$ and $\min(x_j)$ denote the maximum and minimum value of the indicator j in all time points.

Thereafter it is required to calculate the ratio of the index value of the j indicator in year t to the sum of the index value in all years. We use this factor as the basis for the subsequent calculation of information entropy, as shown in the following formula (10). As for positive indexes, the minimum value of the j indicator x_{ij}^* is 0 after the standardization process. Meanwhile, for the negative index, the maximum value of the j indicator x_{ij}^* is 0 after the standardization. In order to avoid that the logarithm is meaningless in the subsequent entropy calculation, the normalized value can be approximated by adding σ to the original basis, σ is an infinitesimal number, this article uses 0.01 to facilitate calculation. Y_{ij} represents the proportion of j indicator in year t .

$$Y_{ij} = \frac{\sigma + x_{ij}^*}{\sum_{i=1}^p (\sigma + x_{ij}^*)} \quad (10)$$

Therefore, the index information entropy value E_j and information utility value D_j (also called information entropy redundancy) can be calculated by the following formulas (11) and (12).

$$E_j = -\frac{1}{\ln p} \sum_{i=1}^p Y_{ij} \ln Y_{ij} \quad (11)$$

$$D_j = 1 - E_j \quad (12)$$

Information entropy redundancy D_j represents the gap between information entropy E_j and 1, and the true information utility value of the

Table 5

Grey correlation degree of environmental regulation and three topic layers in 2 periods.

Sequence 1	Sequence 2	Grey correlation degree (2011–2014)	Grey correlation degree (2015–2018)
Environmental regulation	Economic	0.6696*	0.7686*
	Social	0.6423	0.6918*
	Environment	0.7323*	0.8079*
	Overall	0.6488	0.7483*

Note: The correlation degree of very high is shown as **, and high is shown as *.

Table 6

Simulation comparison between two grey forecasting models.

Year	Actual data sequence (Billion Yuan)	Simulation data 1 (Original model)	Simulation data 2 (improved model)	Relative error 1	Relative error 2
2011	80.20	80.20	80.20	0 %	0 %
2012	108.21	117.05	107.78	8.17 %	0.39 %
2013	137.16	136.13	134.34	0.75 %	2.05 %
2014	156.69	158.33	156.37	1.05 %	0.20 %
2015	186.23	184.15	185.98	1.12 %	0.13 %
2016	229.71	214.18	229.80	6.76 %	0.04 %
2017	244.61	249.10	244.87	1.84 %	0.11 %
2018	285.29	289.72	280.94	1.55 %	1.52 %

observation indicator depends on this value. The larger the D_j , the greater the value of information, the greater the weight of the indicator.

Finally, formula (13) is used to estimate the weight of each indicator. W_j represents the weight of the j^{th} indicator, n represents number of indicators.

$$W_j = \frac{D_j}{\sum_{j=1}^n D_j} \quad (13)$$

As a result, the weighted summation can be used to calculate the evaluation value of each indicator, as shown in formula (14).

$$V_t = \sum_{j=1}^n X_{ij} = \sum_{j=1}^n W_j \frac{x_{ij}^* + \sigma}{\sum_{i=1}^p (\sigma + \frac{x_{ij}^*}{r_i})} * 100 \quad (14)$$

V_t is the comprehensive evaluation value of t year of the sample. X_{ij} is the topic layers value j of time t . To facilitate comparison and calculation, the evaluation value is multiplied by 100, that is, the degree of sustainability performance of the logistics industry in each year. The larger the value, the higher the degree of sustainability performance.

3.2. Methods to calculate grey relation degree

Grey relational analysis is used to judge whether the connection between different sequences is close according to the similarity of sequence curve geometry. The basic method observes the discrete behavior of system factors through linear interpolation converted to a segmented continuous polyline. The closer the polyline geometry, the greater the correlation of the corresponding sequence.

Since the linear assumption does not correspond to the real development trend of the Internet of Things, a principal component analysis and multivariate statistical approaches are not qualified to study the elements influencing the IoT's development (Li et al., 2016). Grey relational analysis is a multifactor method for evaluating the impacts between factors, it is typically used in grey conditions where interactions between factors cannot be described mathematically (Kuo et al., 2008). The IoT development influenced by social-economic-environmental activities is a multifactor system in conformity with the grey characteristics (You et al., 2017; Wang et al., 2019). Besides, due to the limited sources of statistical data in China, coupled with the influence of changes in the consistency of statistic specifications, many data series fluctuate greatly and it is difficult to find typical distribution patterns (Javed et al., 2019). The grey relational analysis makes up for the shortcomings of mathematical statistics. It has low computational complexity and is applicable to any size or characteristics of the samples (Dejamkhooy et al., 2017). Usually, there is little discrepancy between the quantitative results and the qualitative analysis results. Therefore, we conclude it is feasible for the analysis of this paper.

The research on the grey relational model has made great progress in recent decades. Based on the early grey relational model constructed by Deng (1984), Liu (1991) proposed models of absolute correlation,

relative correlation and comprehensive correlation. After that, two grey relational models were constructed based on the similarity and proximity perspectives (Liu et al., 2010). Wei and Xie (2019) then described the rigorous mathematical definitions of positive definiteness, parallelism, consistency, affine, and parallel order preservation of grey correlation degree.

When the dimensions of the sequences $X_i = (x_i(1), x_i(2), \dots, x_i(t))$ and $X_j = (x_j(1), x_j(2), \dots, x_j(t))$ are different, the gray relative correlation degree is usually used to judge their correlation. First, X_{ij} can be rewritten as $X_j = (x_j(1), x_j(2), \dots, x_j(t))$. Elements in sequence X_i and X_j represents the IoT market scale and topic layers value of the logistics industry at time t , respectively. If the two sequences have the same length, the first step is to calculate their initial value sequences, as shown in formula (15).

$$X'_i = \left(1, \frac{x_i(2)}{x_i(1)}, \frac{x_i(t)}{x_i(1)} \right) \quad (15)$$

$$X'_j = \left(1, \frac{x_j(2)}{x_j(1)}, \frac{x_j(t)}{x_j(1)} \right).$$

Then set zero value as the starting point of the two sequences, as shown in (16) and (17).

$$X'_i{}^0 = (X'_i{}^0(1), X'_i{}^0(2), X'_i{}^0(t)) \quad (16)$$

$$x'_i{}^0(t) = \frac{x_i(t)}{x_i(1)} - 1 \quad (17)$$

Next step the $|s'_i|$, $|s'_j|$, $|s'_i - s'_j|$ will be calculated as formula (18)-(20).

$$|s'_i| = \left| \sum_{k=2}^{t-1} x'_i{}^0(k) + \frac{1}{2} x'_i{}^0(t) \right| \quad (18)$$

$$|s'_j| = \left| \sum_{k=2}^{t-1} x'_j{}^0(k) + \frac{1}{2} x'_j{}^0(t) \right| \quad (19)$$

$$|s'_i - s'_j| = \left| \sum_{k=2}^{t-1} (x'_i{}^0(k) - x'_j{}^0(k)) + \frac{1}{2} (x'_i{}^0(t) - x'_j{}^0(t)) \right| \quad (20)$$

Finally, the relative gray correlation degree r_{ij} can be calculated as formula (21). It has the following properties:

$$r_{ij} = \frac{1 + |s'_i| + |s'_j|}{1 + |s'_i| + |s'_j| + |s'_i - s'_j|} \quad (21)$$

- (1) $0 < r_{ij} \leq 1$.
- (2) r_{ij} is only related to the rate of change of the sequence X_i and X_j relative to the initial point, and has nothing to do with the size of each observation value. In other words, multiplication does not change the value of the relative degree of association.
- (3) The closer the change rate of sequence X and J relative to the initial point is, the larger r_{ij} is.
- (4) With different values of r_{ij} , the degree of correlation is as follows: $0 < r_{ij} \leq 0.35$ means that the correlation between sequences is weak. $0.35 < r_{ij} \leq 0.65$ indicates that the correlation between sequences is general. $0.65 < r_{ij} \leq 0.85$ indicates that the correlation between sequences is high. $0.85 < r_{ij} \leq 1$ means that the correlation between sequences is very high.

Therefore, the correlation between the development of the IoT market and the sustainability performance (weighted TBL dimensions) of the logistics industry can be judged by the value of the grey correlation degree r_{ij} . In addition, the relationship between the IoT market and each sub-indicator can be further obtained by same approach. However, some scholars (Guo et al., 2017; Li and Ramanathan, 2018) argued that policy factors like environmental regulation may have an influence on the sustainability performance. In order to verify the role of

the technological innovation of the IoT, we also analyze the correlation between environmental regulation and the sustainable development (also for all weighted TBL dimensions) of the logistics industry. For the period 2011–2018, we select the environmental policy stringency index to measure the pressure induced by environmental regulation (OECD Organisation for Economic Co-operation and Development, 2022). It is a country-specific and internationally-comparable measure of the stringency of environmental policy. Stringency is defined as the degree to which environmental policies put an explicit or implicit price on polluting or environmentally harmful behavior. See Appendix C (OECD Organisation for Economic Co-operation and Development, 2022). See Appendix C (OECD Organisation for Economic Co-operation and Development, 2022).

3.3. Forecast the IoT market using the grey forecasting model

Commonly used quantitative analysis methods of technological innovation forecasting include analogy forecasting, the use of growth curves, and the trend extrapolation method. However, they are not suitable for prediction when encountering uncertain systems with incomplete and inaccurate information. The Grey Model (GM) developed by fuzzy mathematics theory can extract valuable information through the generation and development of existing information in order to correctly characterize the operation behavior of the system and make an accurate forecast of the future (Deng, 1984). GM is generally expressed as GM (n, x), which means: use n -order differential equations to establish a model with x variables. By treating the discrete data scattered on the time axis as a set of continuously changing sequences, using accumulation and subtraction methods, the unknown factors in the grey system are weakened. Finally, a continuous differential equation with time as a variable is constructed, where the parameters are determined by mathematical methods.

As an important category of the GM (n, x), the GM (1, 1) mainly solves the generated sequence with exponential change rule and describes the monotonous change process, such as the development of emerging technologies. Current GM (1, 1) models mainly have four basic forms, including the even GM (1, 1) model (EGM), the original difference GM (1, 1) model (ODGM), the even difference GM (1, 1) model (EDGM) and the discrete GM (1, 1) model (DGM). ODGM, EDGM and DGM can accurately simulate homogeneous exponential sequences for their time response equations. They are all equal-ratio sequences. EGM is most suitable for modeling non-exponential growth sequences and oscillating sequences (Liu et al., 2014). Since actual data is normally not a simple homogeneous exponential sequence or approximate sequence, it is usually inclined to choose EGM in the modeling of uncertain systems with few data samples. Here we used the 1-AGO sequence (accumulated generating operation) generated from the original IoT market data sequence as the only endogenous independent variable for grey forecasting.

As an emerging technology, the IoT is not only under the impact of its endogenous factors, but also influenced by exogenous factors, such as investment confidence, which is usually influenced by current business environment (Ding et al., 2019). As for logistics industry, the Logistics Prosperity Index (LPI) released in 2013 added new perspectives for observing, predicting, and analyzing the development of China's logistics industry, providing a basis for guiding enterprise production, operation and investment activities (CLY, 2019). Therefore, we select the LPI to represent investment confidence. The LPI is a composite index weighted by five indexes: total business volume, new orders, employees, inventory turnover, and equipment utilization.⁵ It reflects the overall changes in the economic development of the logistics industry, with 50

⁵ It adopts the PPS (Probability Proportional to Size) sampling and the number of samples is determined by the contribution of each related industry to the business income of logistics owners.

% as the demarcation point of economic vitality. The LPI higher than 50 % reflects the economic expansion of the logistics industry. Therefore, the relative LPI can be obtained by LPI-50 %. This indicator can guide enterprise production, operation and investment confidence. LPI dataset from 2012 to 2019 could be acquired through China Federation of Logistics & Purchasing (CFLP, 2021), see Appendix C.

Cumulative generation, inverse cumulative generation and grey modeling are the three steps of GM. It uses cumulatively generated operations to establish differential equations.

EGM is summarized as follow steps:

Step 1: Generate 1-AGO sequence $X^{(1)} = (x^{(1)}(1), x^{(1)}(2), \dots, x^{(1)}(t))$ of initial sequence $X^{(0)} = (x^{(0)}(1), x^{(0)}(2), \dots, x^{(0)}(t))$, where $x^{(0)}(t)$ is the time series data at time t , $X^{(1)}$ and $X^{(0)}$ satisfy the following relationship (22).

$$x^{(1)}(k) = \sum_{i=1}^k x^{(0)}(i), k = 1, 2, \dots, t \quad (22)$$

Step 2: Generate mean sequence $Z^{(1)} = (z^{(1)}(2), z^{(1)}(3), \dots, z^{(1)}(t))$, where $z^{(1)}(k)$ and $x^{(1)}(k)$ satisfy the following relationship in formula (23).

$$z^{(1)}(k) = \frac{1}{2}(x^{(1)}(k) + x^{(1)}(k-1)) \quad (23)$$

Step 3: The even value form of the GM (1, 1) can be expressed as Eq. (24), the parameter a represents the development coefficient, which reflects the development trend of $\hat{x}^{(1)}$ and $\hat{x}^{(0)}$, b represents the grey effect.

$$x^{(0)}(k) + ax^{(1)}(k) = b \quad (24)$$

$$\frac{dx^{(1)}}{dt} + ax^{(1)} = b$$

The parameter vector $\hat{a} = [a, b]^T$ in the formula can be estimated by the least square method, as shown in formula (25).

$$\hat{a} = (B^T B)^{-1} B^T Y \quad (25)$$

$$Y = \begin{bmatrix} x^{(0)}(2) \\ x^{(0)}(3) \\ \vdots \\ x^{(0)}(t) \end{bmatrix}$$

$$B = \begin{bmatrix} -z^{(1)}(2) & 1 \\ -z^{(1)}(3) & 1 \\ \vdots & \vdots \\ -z^{(1)}(t) & 1 \end{bmatrix}$$

Step 4: By solving the whitening differential equation of Eq. (24), Eq. (26) can be obtained as follows.

$$x^{(1)}(t) = Ce^{-at} + \frac{b}{a} \quad (26)$$

When $t = 1$, $x^{(1)}(1) = x^{(0)}(1)$, then put it into Eq. (27) to get $C = [x^{(0)}(1) - \frac{b}{a}]e^a$. Therefore, the time response equations of $\hat{x}^{(1)}(k)$ and $\hat{x}^{(0)}(k)$ can be obtained successively, as shown in the following formulas (27) and (28), which make it possible to predict and evaluate the future changes of the IoT market.

$$\hat{x}^{(1)}(k) = \left(x^{(0)}(1) - \frac{b}{a}\right)e^{-a(k-1)} + \frac{b}{a} \quad (27)$$

$$\hat{x}^{(0)}(k) = (1 - e^a) \left(x^{(0)}(1) - \frac{b}{a}\right)e^{-a(k-1)} \quad (28)$$

Step 5: Finally, on the basis of trend changes, the influence of investment confidence needs to be added to obtain the final forecast

sequence, as shown in Eq. (29). $g(k)$ represents the relative LPI at year k .

$$\hat{x}(k) = \hat{x}^{(0)}(k)(1 + g(k))\mu, \mu > 0 \quad (29)$$

In order to ensure the accuracy of the fitting, the objective function containing parameter μ is constructed with the minimum average relative error. It can be easily obtained using fminsearch function of MATLAB.⁶

$$\min \sum_{k=2}^t \frac{|\hat{x}(k) - x^{(0)}(k)|}{x^{(0)}(k)} \times 100\% \quad (30)$$

Step 6: The relative simulation error can be obtained by comparing the simulated data with the actual data, and then the average relative error can be calculated, as shown in formula (31).

$$\varepsilon = \frac{1}{t} \sum_{k=1}^t \frac{|\hat{x}(k) - x^{(0)}(k)|}{x^{(0)}(k)} \quad (31)$$

4. Discussion and analysis

4.1. Results and analysis of sustainability performance of the logistics industry

According to formulas (4)-(7), the time entropy weight between 2011 and 2018 were calculated through fmincon function of MATLAB, as shown in Table 2, the nearest time point has a higher time weight.

Therefore, the weights of each topic and sub-indicator layers of the evaluation framework have been obtained through (8) – (13), as shown in Table 3. The results show that CI contributes to the largest proportion of sustainability performance evaluation. It potentially emphasizes that improving energy consumption efficiency and decreasing greenhouse gas emissions are important targets for a green industry. PI is the indicator with the lowest weight, with around 10 % of the total weight. While other 7 indicators share similar proportion of around 11 %.

Fig. 2 highlights the corresponding trends of each indicator from 2011 to 2018. The overall evaluation value increases from 9.36 to 16.80. The economic and social value reveal a relative steady growth rate of 9.97 % and 10.41 % respectively over this period. In contrast, the environmental dimension hasn't shown an upward change until 2015 and only increased by 5.40 % annually from 2011 to 2018. The strongest contributing indicator for environmental performance has been CI (It has been converted to positive value in models), which grew by 16.40 % on average after 2015.

However, a growth trend of CGP cannot be observed for this period. There is also solely a weak contribution to the improvements of environmental dimension, which is probably led by the hysteresis effect⁷ of patents in accelerating environmental improvement (Kang et al., 2021).

Other economic and social sub-indicators have shown a steady upward trend and contributed to the sustainability performance of the logistics industry. It is worth noting that though TI has risen, the FDI has shown a downward trend after 2015 (See in Appendix B1). The quantitative easing policies of advanced economies (Marcin and Grzegorz, 2020),⁸ the “re-industrialization” strategy in some countries (Wu and Zhang, 2013) and the overall rise in China's domestic labor, land, and environmental costs, and the appreciation of the RMB exchange rate in the past few years have weakened the logistics industry's ability to

⁶ Fminsearch is used in MATLAB to find the minimum value of a multivariate unconstrained function, using the simplex algorithm for calculation.

⁷ Effects of current R&D expenditure and patents on performance are not instant and sometimes take several years to become apparent.

⁸ Although the quantitative easing policies implemented by major central banks have had a positive impact on global asset prices, these plans tend to seriously weaken the international price competitiveness of other economies due to large capital inflows and currency appreciation after quantitative easing abroad.

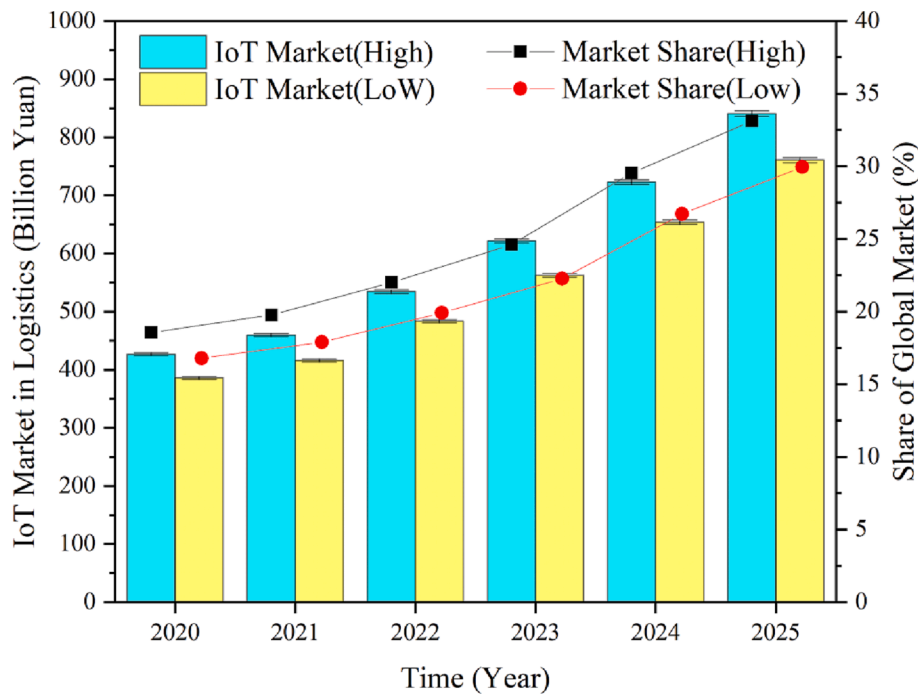


Fig. 3. IoT market forecasting and share between 2021 and 2025 for two scenarios.

Table A1

Logistics industry upgrading evaluation value between 2011 and 2018.

Year	VA	EP	LC	PI	TI	CGP	CP	CI	LP
2011	0.97	0.91	0.91	0.77	0.89	1.73	1.08	1.23	0.94
2012	1.07	1.01	1.04	0.89	0.97	1.15	0.96	1.16	1.05
2013	1.19	1.17	1.15	1.03	1.10	1.19	1.00	1.23	1.16
2014	1.32	1.32	1.26	1.20	1.26	1.42	1.24	1.50	1.28
2015	1.42	1.37	1.38	1.29	1.42	1.06	1.48	1.48	1.40
2016	1.56	1.47	1.49	1.41	1.52	1.58	1.75	1.79	1.54
2017	1.77	1.63	1.63	1.44	1.71	2.13	1.92	2.13	1.76
2018	1.93	1.83	1.81	1.54	1.77	1.85	1.85	2.31	1.87

Table A2

Grey correlation degree of the IoT market and three indicator layers of environment.

Year	PI	CI	CGP
2011–2014	0.7040	0.6358	0.8046
2015–2018	0.8144	0.9623	0.7784

Table A3

Grey correlation degree of the IoT market and environmental regulation.

Year	Correlation Value
2011–2014	0.9357
2015–2018	0.7148

attract FDI (Marcin and Grzegorz, 2020) despite the fact that the proportion of FDI on the logistics sector is quite limited compared to IFI.

4.2. Result and analysis for grey correlation degree

The grey correlation degree compares the relationship between market size of the IoT in logistics industry and the three topic layers. In order to compare the dynamics of correlation values over time and

explore their potential reasons, the original sequence is divided into two equal sub-sequences (2011–2014 and 2015–2018), as shown in Table 4. Similarly, the relationships between environmental regulation and the three topic layers in 2 periods are also listed in Table 5.

The results show that the overall correlation degree between the IoT market and the sustainability performance of the logistics industry in both periods are high, with an upward trend from 0.6708 to 0.9326. In particular, the environmental layer, which reveals the lowest correlation value in the first period reveals the highest correlation value over the second period. As Appendix A2 shows, the correlation between CI and the IoT market has increased to 0.9623. Taking into consideration the development of CI, see A1 in the Appendix, we find that reduction in CI of the logistics industry is also highly correlated with the expansion of the IoT market. A similar increasing trend has also occurred for PI. Logistics as a PM 2.5-intensive activity holds the potential to reduce the PM2.5 intensity with the help of IoT monitoring capability (Zhang et al., 2021). On the contrary, the correlation between the IoT market and CGP is decreasing, which may depict that the application of green technologies has little relationship with the expansion of the IoT in the logistics industry market.

Furthermore, the social layer has the highest correlation with IoT in the first period, but only increased slightly in the second period, though

Table B1

Original data from CLY (2012–2019), CSY (2012–2019) and MEIC (Li et al., 2017; Zheng et al., 2018).

Year	2011	2012	2013	2014	2015	2016	2017	2018
VA(Billion Yuan)	2184.2	2376.3	2604.3	2850.1	3048.8	3305.9	3717.3	4033.7
EP(Million People)	11.62	11.87	12.31	12.74	12.86	13.14	13.56	14.11
LC(Yuan)	47,078	53,391	57,993	63,416	68,822	73,650	80,225	88,508
CGP(‰)	8.84	5.83	6.06	7.23	5.40	8.07	10.86	9.43
PI(g/Yuan)	0.24	0.223	0.203	0.179	0.166	0.150	0.146	0.131
TI(Billion Yuan)	2749.45	3066.14	3588.08	4251.66	4870.66	5295.25	6050.13	6294.23
CP	0.260	0.253	0.255	0.269	0.282	0.298	0.307	0.303
CI(kg/Yuan)	0.285	0.289	0.285	0.271	0.271	0.255	0.237	0.227
LP(Yuan/People)	18.80	20.02	21.15	22.37	23.70	25.16	27.41	28.58

Table B2

Original data from CEIS (Billion Yuan).

Year	2011	2012	2013	2014	2015	2016	2017	2018
IoT Market	258.1	365.0	489.6	600.0	750.1	930.0	1150.2	1430.0

Table B3

Global proportion of the IoT market size in the logistics industry (Wegner, 2021).

Year	2011	2012	2013	2014	2015	2016	2017	2018
Proportion (%)	31.07	29.65	28.02	26.11	24.83	24.7	21.27	19.95

still being significant. One reason could be that China's logistics industry is dominated by small companies. The number of Class A⁹ companies with high income level and high qualification in human capital accounted for only 1.3 % (CLY, 2019). Due to the shortage of funds, insufficient talents, unreasonable resource allocation, and poor management, most small and medium-sized enterprises (SMEs) cannot utilize a full set of integrated and information-based IoT, leading to a situation in which investment flows predominately in a small number of large logistics companies (Lee and Lee, 2015). It has also negatively affected the cooperation between enterprises with different scales on an equal principal. Therefore, it weakened the potential for employment and compensation expansion of these SMEs, which has become a shortcoming in achieving sustainability performance.

Compared with the IoT market size, environmental regulation has a relative lower effect on promoting the sustainable development of the logistics industry. Nevertheless, the correlation values of environmental regulation with all TBL dimensions have been growing. Only for the period 2011–2014, the environmental regulation has had a higher correlation value with the environmental dimension than the logistics industry when China's IoT market size at that time was small. Since the IoT market grew fast, the scale effect amplified the contribution of technological innovation to sustainable development under TBL dimensions. In addition, SMEs, which account for the majority of the market, encountered challenges to hire more employees and increase their profitability under stringent environmental regulations due to their limited financial and investment capabilities (Lee and Lee, 2015). Compared to the advantages (such as efficiency improvements, etc.) of adopting new technologies, the effect of environmental regulation on the social and economic dimensions appears to be relatively low.

However, the growth of the use of IoT technology may have been at least in part driven by more stringent environmental regulation. For this, we conduct an additional grey correlation analysis on the IoT market and environmental regulation, and find that the correlation value drops from 0.9357 in 2011–2014 to 0.7148 in 2015–2018, see Appendix T A3. It indicates that environmental regulation is likely a

Table C1

Environmental Policy Stringency index in China (OECD Organisation for Economic Co-operation and Development, 2022).

Year	2011	2012	2013	2014	2015	2016	2017	2018
Value	1.47	2.47	2.53	2.75	2.89	2.78	2.83	2.86

relevant promoter of the development of IoT technology at an early stage. However, the impact of this policy factor has gradually decreased with the expansion of the IoT market. It depicts that environmental regulation has an indirect positive effect on the sustainability of the logistics industry through support for technological innovation. A too stringent environmental regulation may result in unaffordable prices on environmentally harmful behavior for SMEs, thus reducing their capabilities to adjust investments and even leading to stranded assets (Guo et al., 2017).

4.3. IoT market forecasting under possible scenarios

Based on the actual IoT market sequence between 2011 and 2018, the possible IoT market in logistics capacity in the near future can be estimated through the improved grey forecasting model, Table 6 illustrates the relative data error from the actual data sequence of both original and improved grey model.

The development coefficient a calculated by formulas (24) and (25) is -0.151 , the grey effect b is 96.316 , the average relative error of the simulation data of the original EGM model is 2.66% , the variance obtained by multiplying the sequence by 100 is 9.25 , and the sum of the absolute values of the series residuals is 38.04 . However, the improved EGM model used in this paper has a lower average relative error of 0.56% , the variance obtained by multiplying the sequence by 100 is 0.61 , and the sum of the absolute residuals is only 8.52 , μ is calculated as 0.9507 . Obviously, the error is smaller and the simulation result is more accurate.

The improved EGM model in this article needs to obtain the data of two modules before trend prediction. The first is the prediction data of the original EGM model, and the second is the expected LPI data in the future as the correction parameter for forecasting, which is estimated

⁹ The standard of A-level logistics enterprises is given according to China standard GB/T 19680–2013, their annual logistics operating incomes are over 3 million RMB.

Table C2

LPI dataset from 2012 to 2019 in China (CFLP, 2020).

Year	2012	2013	2014	2015	2016	2017	2018	2019
LPI (%)	54.78	53.03	54.97	55.05	55.23	55.3	53.58	53.47

based on the previous LPI level. However, in 2020, China's logistics industry was strongly influenced by the COVID-19 pandemic, especially the logistics activities in February and March were severely blocked. The LPI fell sharply to 26.2 % compared with the average level of previous years. Then, after the epidemic impacts within the territory of China gradually being controlled, the LPI rebounded significantly. As of December, the average LPI has reached 51.7 %. With the adjusted epidemic response policy, it can be speculated that the annual LPI will likely exceed 45 % in following years.

According to data from previous years and the pandemic influence in China, it can be deduced that the LPI will not likely be less than 45 % in the next five years. However, it will also be difficult to exceed 55 % under the constraints of increased logistics risks caused by the continued global epidemic and trade conflicts (CFLP, 2021). These figures are used as the low and high scenarios of the forecast model, respectively. At the same time, the average relative error of 0.56 % of the improved EGM model is selected as error bar to estimate the range of forecast error. The results are shown in Fig. 3.

According to the forecasting of the model, under the two extreme scenarios, China's IoT market scale in logistics will still maintain a high growth rate of up to 20–25 % each year, which is much higher than China's gradual slowing GDP growth rate (Around 5 %). The market size in both scenarios will eventually exceed 700 billion yuan in 2025. Under two different scenarios, there will be an estimated market size difference of up to 90 billion yuan in 2025.¹⁰ However, considering that main markets are still suffering from COVID-19, GSMA lowered its previous forecast on the scale of the global IoT market in 2025 by 200 billion US dollars (GSMA, 2022). The latest update of IDC's Global IoT Spending Guide shows that IoT spending has increased by 8.2 % in 2020, which is lower than the 14.9 % forecast released in November 2019. Nonetheless, IDC expects global IoT spending to return to double-digit growth rates in 2021 and achieve a compound annual growth rate of 11.3 % during the 2020–2024 forecast period. Considering the forecast data on the global IoT market size and the share of logistics sector (Crook and Vesset, 2020; Wegner, 2021), the Chinese IoT market will likely reach over 30 % of the global market share of logistics in the next five years, which represent a dominant position in the world.

5. Conclusion and outlook

We investigate the influence of increased applications of the IoT technology within the Chinese logistics industry. For the environmental dimension, CI reveals to be a significant indicator to assess the performance of the logistics industry. Hence, from a governmental perspective supportive strategies such as eliminating outdated production capacity, adjusting carbon taxation to improve energy utilization structure or guiding cash flows to high value-added nodes to foster overall industrial sustainability might decrease environmental pressure. Furthermore, combined with our research results, positive social effect could be enhanced by subsidizing IoT technologies with slack resources for small and medium-sized enterprises.

The analysis has shown that environmental related policy regulations can play a positive role in accelerating sustainability under TBL dimensions, but its measurable effect is lower than the effect of the growing IoT market in the long run. Our results suggest that

environmental regulation initially promoted technical innovation, but this impact gradually decreased with the expansion of the IoT market, making IoT implementation a more important driver for improved TBL performance. It is likely that the scale effect of IoT market will be the primary contributor to the TBL performance of China's logistics sector in the future. This follows from our finding that the size of the IoT market correlates well with improved performance on all TBL dimensions of the logistics sector. The likely causality is that IoT supports optimization of the logistic process, a more efficient use of resources, and reduces waste. In total, these observations explain why IoT market has a high positive correlation with the decline in carbon and PM 2.5 emissions. However, the correlation between IoT market and the economic and social dimensions is not so significant. The high investments required to implement IoT systems may limit the creation of additional value added, particularly for SMEs that have limited investment resources (Lee and Lee, 2015). Automation fostered by IoT implementation may also lead to a reduced need for jobs and reduce the compensation to employees.

Based on past trends and LPI data, the global IoT market will likely double within 5 years while the growth rate of China's IoT market will likewise be higher than the world average in the short term. Its share might reach nearly 30 % of the world market in 2025. The huge potential local market size gives Chinese technology companies a highly competitive position. This is also because they have build up a large user base to enhancing their understanding and the output of IoT-based applications (Nir, 2017). In comparison competitors in smaller economies might face severe problems. In order to have access to similar sized markets they need to coordinate multiple legal frameworks, as for instance in the EU. Additionally, they might face several non-obvious dilemmas such as inconsistency of technical standards, variation across infrastructure and specifications for the IoT. A multi-regional market might also require high initial investment, which can become obstacles restricting its large-scale application. Multiple different frameworks might negatively impact the security of logistics information. Technical issues such as chip adaptability and heterogeneous antenna reception systems provide challenges (Roy and Roy, 2019). Furthermore, government's guarantee mechanism and policies for IoT applications can still be incomplete. These issues are potential barriers that limit the spread of IoT technology in some regions. To exploit the potential of IoT for contributing to global environmental goals, at least a transnational green IoT standard for monitoring carbon emissions in production network should be prioritized (Ali and Shahram, 2020).

The influence of COVID-19 is mainly reflected in two regards: the first is that the spread of panic has led to the frustration of investor confidence, which triggers turmoil in the financial and capital markets; the second is that the quarantine measures to control the epidemic have pressed the pause button on the economy, thereby affecting the consumer side (Donthu and Gustafsson, 2020). The lower levels of the production and consumption seriously affected corporate profits and employee compensation, which negatively impacted the diffusion of IoT technology. The strict transportation and population flow control measures in China will not only affect the industrial production, but the integrated global industrial chain will also face challenges (Guan et al., 2020). They at the same time limit the construction of a transnational IoT, which is detrimental to cross-regional collaborative environmental monitoring.

But on the other side, facing the global pandemic, IoT technology can also be regarded as an opportunity to improve the resilience of supply chains (Al-Talib et al., 2020). It can encourage remote cooperation

¹⁰ In the past 5 years, the exchange rate of US dollar against RMB is 6.67 on average.

between partners, and increase the visibility of the logistics system by monitoring supply chain events and patterns in real time (Moayad et al., 2020). Despite the fact that IoT presents both opportunities and challenges for industrial sustainability, supportive regulations are required to ensure that potentials are realized.

Our research has some limitations. First, (grey) correlation analyses as we conduct cannot really prove causation, even though we try to give above some plausible explanations for our observations. Further, we only consider a limited number of factors explaining the TBL performance of the logistics sector, such as the size of the IoT market and the OECD environmental stringency index. It may be because we have overlooked other explanatory factors. These could be identified by different research approaches, such as in-depth practical case studies at company and supply chain level that look at drivers for improved TBL performance.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A

Tables A1–A3.

Appendix B

Tables B1–B3.

Appendix C

Tables C1 and C2.

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